

Marios Vardalachakis

AI/ML Engineer | Software Engineer | Privacy-Preserving ML Systems
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Professional Profile

Software Engineer and AI professional specializing in scalable artificial intelligence infrastructure, privacy-compliant ML systems, and secure full-stack software solutions. Experienced in developing deployable, explainable, and trustworthy AI architectures using Python, FastAPI, PostgreSQL, R, and JavaScript. Track record of designing modular RESTful applications, optimizing secure data pipelines, and bridging performance with strict compliance and security standards. Creator of InvisseeAI and ShinyAnonymizer, two healthcare data anonymization platforms built during a multi-year research career in privacy-preserving ML. InvisseeAI itself was funded through a Horizon Europe submission (SECURED Open Call), written and led as Project Lead.

Education

MPhil by Research in Intelligent Systems (Fully Remote)

University of the Aegean – Department of Cultural Technology and Communication
Mytilene, Greece | 2024–2026
Grade: 9.50/10

M.Sc. in Software Engineering (Fully Remote)

Hellenic Mediterranean University – Department of Electrical and Computer Engineering
Heraklion, Crete, Greece | 2020–2024
Grade: 7.50/10

B.Sc. in Informatics Engineering

Hellenic Mediterranean University – Department of Electrical and Computer Engineering
Heraklion, Crete, Greece | 2011–2020
Grade: 6.31/10

Technical & Research Skills

Artificial Intelligence & Machine Learning: Generative AI & LLM Integration, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Workflow Orchestration, Privacy-Preserving Machine Learning, Differential Privacy & Federated Learning, Trustworthy & Explainable AI, Machine Learning Evaluation Pipelines, PyTorch, TensorFlow, Scikit-learn.

Backend & API Development: Python, FastAPI, RESTful API Development, Asynchronous Backend Architectures, AI Backend Systems, MLOps Workflows, AI Service Integration, Secure AI Infrastructure, Modular Backend Design, Swagger / OpenAPI, Git & Version Control Workflows.

Databases & Data Engineering: PostgreSQL, MySQL, Secure Data Pipelines, Data Anonymization Workflows, Structured Healthcare Datasets.

Data Visualization & Frontend: HTML5, CSS3, JavaScript, D3.js, DC.js, Crossfilter.js, Plotly, Dash (Python), Shiny (R), Interactive Dashboards & Graph Analytics.

Publications

Publications in Journals

- M. Vardalachakis, N. Papadakis, M. Tampouratzis, “ShinyAnonymizer enhanced version and beyond: A further exploration of Privacy Preserving Solutions in Health Data Management”, Applied Sciences Journal, Section: Computing and Artificial Intelligence, Special Issue: Data Visualization Techniques: Advantages & Applications. <https://doi.org/10.3390/app14166921>.

Publications in Conferences

- M. Vardalachakis, M. Vasilakis, and M. Tampouratzis, “A Privacy-Aware Machine Learning Model for IoT Ecosystems,” 2026 7th International Conference in Electronic Engineering & Information Technology (EEITE), 2026. <https://doi.org/10.1109/EEITE69609.2026.11679641>
- M. Vasilakis, A. Peratikou, M. Vardalachakis, M. Tampouratzis, N. Tampouratzis, “An Integrated Security Framework for the Application Lifecycle using SBOM, Wazuh and DefectDojo,” 2026 7th International Conference in Electronic Engineering & Information Technology (EEITE), 2026. <https://doi.org/10.1109/EEITE69609.2026.11679676>
- M. Vardalachakis, M. Vasilakis, and M. Tampouratzis, “A Comprehensive Analysis of Features, Benefits, Challenges, and Best Practices of Security Information and Event Management (SIEM) Solutions,” Computer Sciences & Mathematics Forum, vol. 12, no. 1, Art. no. 18, 2025. <https://doi.org/10.3390/cmsf2025012018>
- M. Vardalachakis, A. Siatras, and C. Kalloniatis, “InviserAI: Advanced Healthcare Data Anonymization Platform,” 2025 16th International Conference on Information, Intelligence, Systems & Applications (IISA), Mytilene, island of Lesbos, Greece, 2025, pp. 1–7. <https://doi.org/10.1109/IISA66859.2025.11311319>
- M. Vardalachakis and C. Kalloniatis, “The role of Anonymization Techniques in Differential Privacy,” 2025 International Conference on Computer Technology Applications (ICCTA), Vienna, Austria, 2025, pp. 14–18. <https://doi.org/10.1109/ICCTA65425.2025.11166256>
- M. Vasilakis, M. Vardalachakis, and M. Tampouratzis, “Approaches for Anonymization Methods in IoT Preservation Privacy,” 2025 6th International Conference in Electronic Engineering & Information Technology (EEITE), Chania, Greece, 2025, pp. 1–5. <https://doi.org/10.1109/EEITE65381.2025.11166166>
- M. Vardalachakis and M. Tampouratzis, “Privacy Preservation in IoT: Anonymization Methods and Best Practices,” 2024 5th International Conference on Communications, Information, Electronic and Energy Systems (CIEES), Veliko Tarnovo, Bulgaria, 20–22 Jan. 2024. <https://doi.org/10.1109/CIEES62939.2024.10811203>
- M. Vardalachakis, M. Tampouratzis, M. Vasilakis, N. Papadakis, “The Future of Privacy: A Review on AI’s Role in Shaping Data Security,” IEEE 5th International Conference in Electronic Engineering, Information Technology & Education (EEITE 2024), 29–31 May, Chania, Crete, Greece. <https://doi.org/10.1109/EEITE61750.2024.10654397>
- M. Tampouratzis, M. Vasilakis, M. Vardalachakis, M. Raftaki, G. Adamidis, M. Antonidakis, “CircuitLib: A Case Study of Spin-Off e-Business in Embedded and Electronic Systems Development,” IEEE 5th International Conference in Electronic Engineering, Information Technology & Education (EEITE 2024), 29–31 May, Chania, Crete, Greece. <https://doi.org/10.1109/EEITE61750.2024.10654406>
- M. Vardalachakis, H. Kondylakis, M. Tampouratzis, N. Papadakis, N. Mastorakis, “Anonymization, Hashing and Data Encryption Techniques: A Comparative Case Study,” IEEE 3rd International Conference on Applied Mathematics & Computer Science (ICAMCS), Lefkada Island, Greece, August 8–10, 2023. <https://doi.org/10.1109/ICAMCS59110.2023.00028>
- M. Vardalachakis, H. Kondylakis, L. Koumakis, A. Kouroubali, D. G. Katehakis, “ShinyAnonymizer: A Tool for Anonymizing Health Data,” in ICT4AWE. <https://doi.org/10.5220/0007798603250332>

Research & Professional Experience

AI/ML Full-Stack Engineer & Researcher

University of the Aegean – PrivaSI Lab & PRIVACT P.C. | 2024 – 2026

Project: InvisseeAI (Advanced Healthcare Data Anonymization Platform) [GitHub](#) | <https://privact.eu/invisseeai/> | July 2025 – December 2025

- Built InvisseeAI, a platform for anonymizing sensitive healthcare datasets.
- Built the ML pipelines and backend services that connect data processing with the ML/MLOps side of the system.
- Designed the REST APIs and backend architecture that let the ML pipeline talk securely to the healthcare data systems.
- Added privacy-enhancing techniques to the pipeline — differential privacy, federated learning, and secure multiparty computation.
- Contributed AI components to Horizon Europe research projects spanning multiple countries.
- Improved the backend architecture for better scalability and stronger protection of sensitive data.
- Ran experiments to evaluate and validate the privacy-aware ML models.
- Worked on balancing data usefulness against privacy, using techniques like k-anonymity and t-closeness.

Administrative & Secretarial Support Staff

National Athletic Center of Heraklion (EAKH) | 2022 – 2023

- Provided administrative support to management and staff; managed incoming/outgoing correspondence protocol and record-keeping.
- Worked with partner organizations and public services; welcomed and assisted visitors, and provided basic IT support to staff.

R&D Engineer (Internship & Thesis Project)

Foundation for Research and Technology – Hellas (FORTH) | 2016 – 2018

Project: ShinyAnonymizer (A Tool for Anonymizing Health Data) [GitHub](#) | <https://www.ics.forth.gr/cbml/cbml/project/1929?lang=en>

- Built ShinyAnonymizer, an R-based tool that makes anonymizing clinical datasets easier for non-technical researchers.
- Combined hashing, encryption, and anonymization techniques with a PostgreSQL backend to keep the data secure end to end.
- Built dashboards that let researchers configure privacy parameters and see how the data looks in real time.
- Collaborated on the Horizon 2020 project “iManageCancer”, contributing to the development of tools that empower cancer patients through secure data management.

Additional Projects: <https://mariosvard.github.io/mariosvardalachakis>

Projects

Technology InvisseeAI: Advanced Healthcare Data Anonymization Platform

GitHub: <https://github.com/mariosggg/InvisseeAi>

Tech Stack: R, Shiny, PostgreSQL, Plotly

InvisseeAI is a privacy-enhancing platform designed to address the healthcare industry’s challenges in managing sensitive patient data. The platform employs advanced anonymization techniques, including k-anonymity, l-diversity, t-closeness, and differential privacy, to protect patient confidentiality without sacrificing data utility. By integrating secure multiparty computation (SMPC) and federated learning, InvisseeAI enables decentralized data analysis across healthcare institutions while ensuring compliance and data security.

Technology ShinyAnonymizer: A Tool for Anonymizing Health Data

Github: <https://github.com/mariosggg/ShinyAnonymizer>

Tech Stack: R, Shiny, PostgreSQL, Plotly

ShinyAnonymizer is an open-source tool designed to securely anonymize and encrypt sensitive health and medical datasets. It allows non-expert users to connect to remote databases (like PostgreSQL) or flat files (CSV/Excel) and apply rigorous privacy, hashing, and encryption.

Technology RiskSense – Predictive Intelligence for Early Health Monitoring

Github: <https://github.com/mariosvard/RiskSense>

Live Demo: <https://mariosvard.github.io/RiskSense/>

Tech Stack: HTML, CSS, JS, D3.js, DC.js, Crossfilter.js

RiskSense is an interactive analytics dashboard which combines environmental and contextual factors into a dashboard that can be used for early health monitoring. RiskSense combines various weak indicators (such as temperature, humidity, number of tourists, etc.) into one score using a confidence measure.

Technology Sentinel – AI-Assisted Hybrid Threat Intelligence Platform

Github: <https://github.com/mariosvard/Sentinel-AI-Assisted-Hybrid-Threat-Intelligence-Platform>

Live Demo: <https://mariosvard.github.io/Sentinel-AI-Assisted-Hybrid-Threat-Intelligence-Platform/>

Tech Stack: HTML, CSS, JS, D3.js, DC.js, Crossfilter.js

Sentinel is an intelligent analytics platform that helps analysts investigate suspicious online behavior, coordinated narratives, bot amplification, malicious URLs, and hybrid information threats through explainable AI and interactive visual analytics.

Technology IoT Malware Dashboard

Github: https://github.com/mariosggg/IotMalware_Dashboard

Live Demo: https://mariosggg.github.io/IotMalware_Dashboard/

Tech Stack: HTML, CSS, JS, D3.js, DC.js, Crossfilter.js

A fully interactive dashboard for exploring IoT malware incidents, trends, and statistics in real time.

Technology Cybercrime Complaints & Losses Dashboard

Github: <https://github.com/mariosggg/IncidentResponse>

Live Demo: <https://mariosggg.github.io/IncidentResponse/>

Tech Stack: HTML, CSS, JS, D3.js, DC.js, Crossfilter.js

A fully interactive Cybercrime Complaints & Losses Dashboard for real-time data exploration.

Technology Epsilon – Ontology Evaluation System

Github: https://github.com/mariosggg/Epsilon_Ontology_Evaluation_System

Live Demo: https://mariosggg.github.io/Epsilon_Ontology_Evaluation_System/

Tech Stack: HTML, CSS, JavaScript, REST API Integration

An online evaluation system that shows metrics, statistics, and pitfalls of ontologies, integrating with the OOPS REST API to detect common ontology modeling pitfalls.

Research Projects & Funding

InvisseeAI: Healthcare Data Anonymization Platform

SECURED Open Call – Horizon Europe (Grant Agreement No. 101095717) | July 2025 – December 2025

- **Role:** Project Lead / Core Developer
- **Collaborators:** PRIVACT P.C. & PrivaSI Lab, University of the Aegean
- Wrote and submitted the proposal that secured the funding.
- Owned the technical architecture and backend design, and saw the implementation through.
- Built the anonymization workflows, combining differential privacy, federated learning, and secure multiparty computation.
- Presented the work at IISA 2025.

ShinyAnonymizer: Healthcare Data Anonymization Platform

iManageCancer – Horizon 2020

- **Role:** Core Developer
- **Organization:** Computational Biomedicine Lab (CBML), Foundation for Research and Technology – Hellas (FORTH)
- Designed and built the healthcare data anonymization platform.
- Combined anonymization, hashing, and encryption into the system.
- Contributed to Horizon 2020 research deliverables and demonstrations.
- Published research outcomes at ICT4AWE 2019.

Certifications & Training

- **Certificate of Achievement – Power BI** — DigDATable | 21 Feb 2025. Focus: Data visualization, development of analytical dashboards, and data-driven problem solving.
- **Certificate of Completion – Fundamentals of Web Development** — The Prince's Trust International / 100 Mentors / Bizrupt | Dec 2023 – Jan 2024. Focus: Frontend development fundamentals, including HTML, CSS, JavaScript, and AngularJS; emphasis on client-side architecture and user interface development.

Languages

- English – Professional Working Proficiency (B2 Certified). Certification: GoLearn Certificate.

Hackathons & Innovation Activities

- Finalist Team Member, **National Security Innovation Challenge** (2026), organized by the Hellenic National Intelligence Service (EYP), hosted at Stavros Niarchos Foundation Cultural Center. Project: *SentiNeL – AI-assisted Hybrid Threat Investigation & Intelligence Analysis Platform*.
- Selected Team Participant, **Hackathon: Reimagining eHealth in Crete** (2026), hosted at Foundation for Research and Technology – Hellas (FORTH). Project: *RiskSense – Predictive Intelligence for Early Health Monitoring*, an interactive analytics platform for AI-driven health risk assessment using environmental and contextual indicators.

Additional Interests

- Development of personal projects in Artificial Intelligence and Data Analytics
- Engagement with data visualization technologies and interactive applications
- Participation in hackathons and innovation competitions
- Continuous learning in data security and privacy protection
- Music and learning musical instruments, fostering creativity and discipline